	PC	Fundamentals of Signal Processing & Communication	3	1	0	4	4
	PC	Computer Organization & Microprocessor	3	0	0	3	3
	PC	Computer Organization & Microprocessor Laboratory	0	0	2	2	1
			17	**1**	**8**	**26**	**22**
Sem V	OE	Open Elective 2 **((NPTEL/SWAYAM/MOOC))**	3	0	0	3	3
	HSC	Engineering Economics	3	0	0	3	3
	PE	Program Elective 1	3	0	0	3	3
	PC	Theory of Computation & Compiler Design	3	0	0	3	3
	PC	Theory of Computation & Compiler Design Laboratory	0	0	2	2	1
	PC	Digital Signal Processing	3	0	0	3	3
	PC	Digital Signal Processing Laboratory	0	0	2	2	1
	PC	RF Engineering	3	0	0	3	3
	PC	RF Engineering Laboratory	0	0	2	2	1
			18	**0**	**6**	**24**	**21**
Sem VI	OE	Open Elective 3 (From Other Department of FoET)	3	0	0	3	3
	PE	Program Elective 2	3	0	0	3	3
	PE	Program Elective 3	3	0	0	3	3
	PC	Operating Systems	3	0	0	3	3
	PC	Operating Systems Laboratory	0	0	2	2	1
	PC	Digital Communication	3	0	0	3	3
	PC	Digital Communication Laboratory	0	0	2	2	1
	PC	Computer Communication and Networking	3	0	0	3	3
	PC	Computer Communication and Networking Laboratory	0	0	2	2	1
			18	**0**	**6**	**24**	**21**
Sem VII	Pro	Summer Internship	0	0	0	0	2
	OE	Open Elective 4 (From Other Department of FoET)	3	0	0	3	3
	PE	Program Elective 4 (3-0-0)	3	0	0	3	3
	PE	Program Elective 5 (3-0-0)	3	0	0	3	3
	PC	Software Engineering Methodology	3	0	0	3	3
	PC	Software Engineering Methodology Laboratory	0	0	2	2	1
	PC	Embedded Systems	3	0	0	3	3
	PC	Embedded Systems Laboratory	0	0	2	2	1
	PC	Digital CMOS and VLSI Design	3	0	0	3	3
	PC	Digital CMOS and VLSI Design Laboratory	0	0	2	2	1
		Seminar					1
			18	0	6	24	24
Sem VIII		**Course Name**	**Th**	**Tut**	**Pra**	**Hrs**	**Cr**
	Pro	Major/Comprehensive Project					**12**

Program Core Electives:

Semester	Software and Networks	Artificial Intelligence	Communication and Signal Processing	Embedded and VLSI
5 (program core elective1)	Web Technology	Optimization methods and Algorithms		Introduction to CMOS and Memory Technology
	Problem solving through Java			
6 (program core elective2, 3)	Cloud Architecture and Service	Machine Learning	Image Processing	Hardware Accelerated Computing
	Advanced Web Technology	Data warehousing & Mining	Statistical Signal Processing	
	Advanced Algorithm Design			
7 (program core elective4, 5)	Cryptography and Network Security	Computer Vision	Optical Communication	Autonomous systems
	Internet of Things	Deep Learning and Reinforcement learning	Modern Wireless Communications	Hardware Accelerated Computing
	Blockchain Technology			Real time Operating Systems
	Mobile Application Development			

7th Semester

<Course Code>					Software Engineering Methodologies					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

1. To provide the idea of decomposing the given problem into Analysis, Design, Implementation, Testing, and Maintenance phases.
2. To provide an idea of using various process models in the software industry according to given circumstances.
3. To gain knowledge of how Analysis, Design, Implementation, Testing, and Maintenance processes are conducted in a software project.

UNIT I: SOFTWARE PROCESS MODELS	**12 Hrs.**
Software Product, Software crisis, Handling complexity through Abstraction and Decomposition, Overview of software development activities, Process Models, Classical waterfall model, iterative waterfall model, prototyping mode, evolutionary model, spiral model, RAD model, V Model, Agile models: Extreme Programming, and Scrum. Requirement Gathering and Analysis, Functional and Non-functional requirements, Software Requirement Specification (SRS).	
UNIT II: STRUCTURED ANALYSIS & DESIGN	**10 Hrs.**
Overview of design process: High-level and detailed design, Cohesion and coupling, Modularity and layering, Function–oriented software design: Structured Analysis using DFD Structured Design using Structure Chart, Basic concepts of Object-Oriented Analysis & Design. User interface design, Command language, menu and iconic interfaces, UML.	
UNIT III: CODING AND SOFTWARE TESTING TECHNIQUES	**10 Hrs.**
Coding, Code Review, documentation. Testing: - Unit testing, Black-box Testing, Whitebox testing, Cyclomatic complexity measure, coverage analysis, mutation testing, Debugging techniques, Integration testing, System testing, Regression testing.	
UNIT IV: SOFTWARE RELIABILITY AND SOFTWARE MAINTENANCE	**10 Hrs.**
Basic concepts in software reliability, reliability measures, reliability growth modeling, Quality SEI CMM, Characteristics of software maintenance, software reverse engineering, software reengineering, and software reuse.	
	Total: 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Understand and decompose the given project in various phases of a lifecycle.
CO2	:	Apply the appropriate process model depending on the user requirements.
CO3	:	Evaluate the various life cycle activities like Analysis, Design, Implementation, Testing, and Maintenance.
CO4	:	Analyze the various processes used in all the phases of the product
CO5	:	Design the knowledge, techniques, and skills in the development of a software product.
CO6	:	Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions

TEXT/REFERENCE BOOKS

13. Roger S. Pressman "**Software Engineering, A Practitioner's Approach**", TMGHill.
14. I. Sommerville "**Software Engineering**",9th Ed., Pearson education.
15. Rajib Mall "**Fundamentals of Software Engineering**", PHI,2014.

<Course Code>					Software Engineering Methodologies Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

5. To explain the fundamental concepts of Software Engineering Lifecycle models.
6. To understand the software requirement specifications and the SRS documents.
7. To examine the various design and development solutions with proper analysis
8. To demonstrate the software project management skills through case studies.

LIST OF EXPERIMENTS	
1	To study how to define and develop of problem statement
2	To study the preparation of Software Requirement Specification Documents, Design Documents for Quiz System, Online Ticket Reservation System, and Course Registration System
3	To study the DFD Model (Level 0, Level 1 DFD and data dictionary) of the sample problem (Use of a CASE tool required) Develop a structured design for the DFD model developed
4	To draw the Class Diagrams and show Various Class Relationships.
5	To draw the UML Component and Deployment Diagram for the identified System
6	To study the use cases and develop the Use case Model with include and external Relationships
7	To study the identified Scenarios find interaction between objects and represent using Sequence diagram
8	To study the identified Scenarios find interaction between objects and represent using Collaboration diagram
9	To draw the relevant Activity Diagram for the Same System
10	To draw the relevant State Chart diagram for the same System

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Understand the basic concepts of Software Engineering.
CO2	:	Ability to translate end-user requirements into system and software requirements.
CO3	:	Analyze problems, and identify and define the computing requirements appropriate to its solution.
CO4	:	Evaluate various techniques, metrics, and strategies for Testing software projects.
CO5	:	Demonstrate the integration testing which aims to uncover interaction and compatibility problems as early as possible.
CO6	:	Design different case studies on cost estimation of varying software models in a group as a team leader.

TEXT/REFERENCE BOOKS

16. Roger S. Pressman "**Software Engineering, A Practitioner's Approach**", TMGHill.
17. I. Sommerville "**Software Engineering**",9th Ed., Pearson education.
18. Rajib Mall "**Fundamentals of Software Engineering**", PHI,2014.

24ICxxxT					Embedded Systems					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To introduce the foundational principles of embedded systems design and development.
- To acquire proficiency in programming microcontrollers and interfacing them with sensors, actuators, and other peripherals.
- To explore interfacing techniques for sensors, actuators, and other peripherals in embedded systems.
- To design and implement embedded systems solutions for real-world challenges through hands-on projects.

UNIT 1: EMBEDDED SYSTEMS – INTRODUCTION AND ARCHITECTURE	**11 Hrs.**
Introduction to embedded system (ES); Characteristics and functionalities; Classification of Embedded Systems; Microcontroller for Embedded Systems: microcontroller architecture, pin diagram and basic connections – clock, reset, brown-out, etc.; General register set; Special function registers; On-chip memories; General Purpose Inputs and Outputs (GPIOs)	
UNIT 2: ASSEMBLY PROGRAMMING FOR EMBEDDED SYSTEMS	**12 Hrs.**
Instruction Set and Assembly language programming: Data transfer, arithmetic, logic and branch instructions, addressing Modes; Basic programming concepts: Looping, branching, memory-access, time-delays, using assembly instructions; Stack and related operations; Subroutines and interrupts.	
UNIT 3: PERIPHERAL PROGRAMMING USING EMBEDDED C	**12 Hrs.**
Programming the on-chip modules/capabilities of the microcontroller using Embedded C: GPIOs, Timers/Counters, Waveform generation, EEPROM, ADC, Interrupt functionalities, Watchdog timer. Basics of parallel and serial communication; Programming the serial communication modules: USART, Serial Peripheral Interface (SPI), I2C Inter-integrated Circuit (I2C).	
UNIT 4: I/O INTERFACING IN EMBEDDED SYSTEM	**10 Hrs.**
Interfacing of I/O devices, LED and switches, 4x4 Keypad, 16x2 LCD, Motor drivers and opto-couplers, Stepper Motor, Servo Motor, DC Motors, Relays, Buzzers, Proximity sensor, temperature sensor, DAC, RTC. Interfacing wireless modules: Bluetooth, Wi-Fi, Zigbee.	

Total: 45 Hrs

COURSE OUTCOMES

On completion of the course, the student will be able to:

CO1- Describe the basic components/functionalities, and architecture of microcontroller based embedded system.

CO2- Write simple programs using assembly language, and summarize the instruction set architecture for microcontroller.

CO3- Apply the syntax and constructs of the C language, to write Embedded C programs.

CO4- Analyze the functionality of on-chip modules/capabilities of the microcontroller: timers/counters, interrupts, basic communication protocols, EEPROM, ADC, waveform generator.

CO5- Comprehend and demonstrate the working of external input-output devices (interfacing of sensors and actuators) to the microcontroller.

CO6- Develop simple embedded system, utilizing Embedded C programming for interfacing input-output devices with the microcontroller and/or the related on-chip modules.

TEXT/REFERENCE BOOKS

1. Muhammad Ali Mazidi, Sarmad Naimi, and Sepehr Naimi. "AVR Microcontroller and Embedded Systems: Using Assembly and C", (2015), Pearson India.

2. Raj Kamal, "Embedded Systems: Architecture, Programming, and Design", 2nd Edition, Tata McGraw-Hill.

3. Wayne Wolf, "Computers as Components: Principles of Embedded Computing System Design" – (second edition), Morgan Kaufmann

4. K.V. Shibu, "Introduction To Embedded Systems" - 2nd Edition, Tata McGraw-Hill

24ICxxxP					Embedded Systems Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	25	25	50

COURSE OBJECTIVES

- To impart knowledge of embedded system development.
- To familiarize with Microcontroller Boards and IDEs for Embedded System Development.
- To interface peripherals to Embedded Systems.
- To build a small-scale Embedded System.

Laboratory Sessions would be based on following topics but not limited to:

1. Familiarization with IDE and trainer kits/boards.
2. Program for blinking LED pattern generation.
3. Program for interfacing multi-digit 7 segment display and implementing counter.
4. Program for interfacing toggle and push button switches, simple keypad, and matrix keypad.
5. Program for interfacing Buzzer, Relay, DC motor, Servo motor, Stepper motor.
6. Program for interfacing LCD and displaying text on it.
7. Program for interfacing various sensors and displaying quantity on LCD.
8. Program for interfacing speech and voice recognition modules and controlling peripherals.
9. Program for interfacing camera modules.
11. Program for interfacing RS 232 serial modules and file transfer using it.
12. Program for interfacing Ethernet module and transferring files using it.
13. Program for interfacing wireless modules: Zig bee, Bluetooth, GSM modem, GPS module, RFID etc.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Remember the basic components, and architecture of embedded systems.
CO2- Understand programs using assembly language and C programming.
CO3- Apply the concepts of interfacing for embedded systems.
CO4- Analyze different scenarios of embedded systems.
CO5- Evaluate the working of different embedded systems.
CO6- Create a small-scale embedded system for real world applications.

TEXT/REFERENCE BOOKS

1. Shibu K. V, Introduction to Embedded Systems, TMH.
2. Frank Vahid, Tony Givargis, Embedded system design: A unified Hardware/Software introduction, Wiley.
3. Rajkamal, Embedded System: Architecture, Programming and Design, TMH.
4. Wayne Wolf, Morgan, Computer as Components: Principles of Embedded Computing System Design, Kaufmann Publication.
5. Muhammad Ali Mazidi, 8051 Microcontroller and Embedded Systems: Using Assembly and C, Pearson education India.
6. Muhammad Ali Mazidi, AVR Microcontroller and Embedded Systems: Using Assembly And C, Pearson education India.
7. Muhammad Ali Mazidi, PIC Microcontroller and Embedded Systems: Using Assembly And C, Pearson education India.

24ICxxxT					Digital CMOS and VLSI Design					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	0	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To understand the characteristics and concepts related to the design of digital CMOS VLSI circuits/gates.
- To explore various CMOS logic styles, and design CMOS VLSI circuits/gates at the transistor level and layout level.
- To analyze the performance/power of digital CMOS VLSI circuits/gates.

UNIT 1 INTRODUCTION TO DIGITAL CMOS VLSI	08
Trends in VLSI/Semiconductor industry, related to technology (Moore's) scaling; ITRS Roadmap; Overview of semiconductor devices inherent in the MOSFET: MOSFET I/V Characteristics, MOSFET C/V Characteristics, Second Order Effects. DC Transfer Characteristics: CMOS inverter DC Characteristics, Beta Ratio Effects, Noise Margins, Pass Transistor DC Characteristics	
UNIT 2: CIRCUIT CHARACTERIZATION AND PERFORMANCE ESTIMATION	12
RC Delay Model, Linear Delay Model, Logical Effort, Parasitic Delay, Delay in a Logic Gate, Delay in Multistage Logic Networks. Timing analysis delay models **Power:** Sources of power, Dynamic Power, Static Power, Energy-Delay optimization, Low-power design techniques.	
UNIT 3: DESIGN OF DIGITAL CMOS VLSI STANDARD AND COMPOUND GATES	14
Combinational Design: Circuit Families: Pass-transistor tree-based logic gates (and similar other logic styles – CPL, transmission gates, DPL, etc); Pseudo-NMOS logic; CVLS logic; Dynamic logic (domino, NP domino, Zipper); Hybrid logic style and examples (Full-Adder, MUX, XOR), Pitfalls of various families, Stick diagram. Sequential Design: Sequencing Static Circuits, Max-Delay Constraints, Min-Delay Constraints, Time Borrowing, Clock Skew, Circuit Design of Latches and Flip-Flops	
UNIT 4: Datapath and array subsystem design	8
Carry ripple adder, carry look ahead adder, Manchester carry chain adder, carry skip adder, carry select adder, shifters, multipliers, SRAM, DRAM, and other memories.	

Total: 42 Hours

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Identify the pros and cons, and the trends related to scaling of the CMOS VLSI technology.
CO2 - Understand the characteristics, sub-parts, and regions of operation of the MOSFET, and use their I-V equations.
CO3 – Design of the combinational and sequential circuits using digital CMOS VLSI standard gates and compound gates, at the transistor –level.
CO4 - Estimate the performance/power of digital CMOS VLSI standard and compound gates.
CO5 - Analyze and compare various digital CMOS logic styles and gates/circuits.
CO6 - Create layouts of digital CMOS VLSI standard gates and compound gates, based on Lambda rules.

TEXT/REFERENCE BOOKS

- Neil Weste, David Harris, Ayan Banerjee, "CMOS VLSI Design: A circuits and systems perspective", 3rd Edition, Pearson.
- Sung-Mo Kang, Yusuf Leblebici, "CMOS Digital Integrated Circuits", 3rd Edition, Tata McGraw Hill.
- Robert F. Pierret, "Semiconductor Device Fundamentals", 1st Edition, Pearson.Prentice Hall.

XXXX					Digital CMOS and VLSI Design Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	0	0	0	50	50	100

COURSE OBJECTIVES

- To learn how to create layout designs for CMOS circuits. This involves understanding the physical aspects of CMOS, like transistor placement, routing, and metal layers.
- To Use CAD (Computer-Aided Design) tools to simulate and verify the functionality of CMOS designs.
- To Gain proficiency in designing digital integrated circuits using CMOS technology. This includes understanding the basics of CMOS design, such as logic gates, flip-flops, and other digital building blocks.

List of experiments:

1. To derive the CMOS INV gate noise margins using DC analysis.
2. To design layout of the CMOS INV and perform the RCX extraction and measure the delays.
3. To design layout of the CMOS NAND2 and perform the RCX extraction and measure the delays.
4. To design layout of the CMOS NOR2 and perform the RCX extraction and measure the delays.
5. To measure the dynamic and static power consumption of the CMOS NAND gate.
6. To understand the FO4 delay and measure the FO4 delay of the INV.
7. To apply the Logical effort technique to reduce the path delay.
8. To design layout of the Pass transistor AND2 and perform the RCX extraction and measure the delays.
9. To design layout of the DOMINO NAND2 and perform the RCX extraction and measure the delays.
10. To design layout of the Psuedo-NMOS NOR2 and perform the RCX extraction and measure the delays.

COURSE OUTCOMES

On completion of the course, students will be able to

CO1: Identify the pros and cons, and the trends related to VLSI Design.
CO2: Create layout designs for CMOS circuits and measure the performance metrics.
CO3: Understand the physical aspects of CMOS, including transistor placement, routing, and the use of metal layers.
CO4: Use CAD tools to simulate and verify the functionality of CMOS designs.
CO5: Perform static timing analysis to ensure that CMOS circuits meet their timing constraints.
CO6: Use techniques for minimizing power consumption in CMOS circuits. Understand the trade-offs between speed, power, and area.

TEXT/REFERENCE BOOKS

19. Neil Weste, David Harris, Ayan Banerjee, “CMOS VLSI design: A circuits and systems perspective”, 3rd Edition, Pearson.
20. Sung-Mo Kang, Yusuf Leblebici , “CMOS Digital Integrated Circuits”, 3rd Edition, Tata McGraw Hill.

Program Core Elective

7th Semester

24ICxxxT					Cryptography and Network Security					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To understand the concept of security requirements, security attacks, and security policy.
- To understand the mathematical concepts for cryptographic algorithms.
- To be able to secure a message over insecure channel by various means.
- To understand the security analysis of cryptographic algorithms.
- To understand various protocols for network security to protect against the threats in the networks.

UNIT 1: INTRODUCTION AND BACKGROUND	**12 Hrs.**
Introduction to Cryptography, Security Threats, Vulnerability, Active and Passive attacks, Security services and mechanism, Conventional Encryption Model, CIA model, Classical Cryptographic Techniques, Introduction to Modular Arithmetic and Number Theory.	
UNIT 2: SYMMETRIC AND ASYMMETRIC KEY CRYPTOGRAPHY	**10 Hrs.**
Block Ciphers (DES, AES) : Feistal Cipher Structure, Simplifies DES, DES, Double and Triple DES, Block Cipher design Principles, AES, Modes of Operations, Public-Key Cryptography : Principles Of Public-Key Cryptography, RSA Algorithm, Diffie- Hellman Key Exchange.	
UNIT 3: HASH FUNCTION AND DIGITAL SIGNATURE	**10 Hrs.**
Authentication Requirement, Functions, Message Authentication Code, Hash Functions, Security Of Hash Functions And Macs, MD5 Message Digest Algorithm, Secure Hash Algorithm, Digital Signatures, Key Management.	
UNIT 4: SECURITY IN NETWORKS	**10 Hrs.**
Threats in networks, Electronic mail Security, IP Security, Web security, System Security, Firewall.	

Total: 42 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Define the importance of security policies, basic concepts of cryptography, and cryptanalysis
CO2: Apply appropriate encryption techniques to secure data in transit across data networks.
CO3: Analyze cryptography algorithms with the knowledge of security requirements and security attacks
CO4: Evaluate the authentication and hash algorithms as per security requirements.
CO5: To examine the issues and structure of Authentication Service and Electronic Mail Security
CO6: Discuss the security analysis of framework and policies applied in real life applications.

TEXT/REFERENCE BOOKS

21. William Stallings, “Cryptography and Network Security Principles and Practice”, Pearson Education
22. Behrouz A. Forouzan, “Cryptography and Network Security”, McGraw-Hill Education
23. William Stallings , “Network Security Essentials: Applications and Standards”, Prentice Hall
24. W. Mao, “Modern Cryptography – Theory and Practice”, Pearson Education.

<******>					< Optical Communication >					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To introduce the students to various optical fiber modes, configurations, and various signal degradation factors associated with optical fiber
2. To understand various optical sources and optical detectors
3. To understand the optical communication system, optical amplifiers.
4. To study different fiber network elements and basic optical components.

UNIT I: OPTICAL FIBER BASICS	**11 Hrs.**
Introduction to Optical Fiber, Wave Propagation in Step-Index & Graded-Index Fiber, Modes & Rays. Basic Optical Communication System, Advantage and applications of Optical Communication System, Absorption, Scattering, Overall Fiber Dispersion-Single Mode Fibers, Multimode Fibers, Polarization.	
UNIT II: COMPONENTS OF OPTICAL FIBER COMMUNICATION SYSTEMS	**10 Hrs.**
Couplers, Isolators and Circulators, Multiplexers and Filters, Lasers, Light-Emitting Diodes, Photodetectors, Switches, Wavelength Converters.	
UNIT III: OPTICAL AMPLIFIERS	**9 Hrs.**
Stimulated Emission, Spontaneous Emission, Erbium-Doped Fiber Amplifiers, Raman, Semiconductor Optical Amplifiers, Crosstalk in SOAs, Crosstalk and Noise.	
UNIT IV: OPTICAL NETWORKS	**12 Hrs.**
SONET/SDH, Optical Transport Network, Optical Access Network, Ethernet, Network Topologies. Advanced Multiplexing Strategies- OFDM, SDM. Subcarrier Multiplexing, WDM Network Architectures, Few Advanced Topics: Flexible Optical Networks, Optical Neural Network, Cognitive Optical Network.	
TOTAL HOURS: 42 Hrs.	

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Remember the basic concepts of optical fiber structure and optical communication systems.
CO2	:	Understand different design aspects of optical fiber including its advantages and limitations.
CO3	:	Apply optical fiber and optical communication concepts.
CO4	:	Analyze the system performance of optical transmitters, receivers, and optical amplifiers.
CO5	:	Evaluate the design of optical fiber communication systems.
CO6	:	Create optical communication applications in the recent technological trends.

TEXT/REFERENCE BOOKS

1. Gerd Keiser, "**Optical Fiber Communications**", 4th Edition McGraw Hill.
2. John M. Senior, "**Optical Fiber Communication**" PHI/Pearson.
3. G. P. Agrawal, "**Fiber optic Communication Systems**", John Wiley and sons.
4. John Gowar, "**Optical Communication Systems**" Pearson.

<Course Code>					Internet of Things					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

4. To impart knowledge about components of IoT, architecture, and platforms of IoT.
5. To apprise students with basic knowledge of IoT that paves a platform to understand physical, logical design, and business models.
6. To teach a student how to analyse requirements of various industrial systems for cost-effective design of IoT applications on different IoT platforms.
7. To explain the students how to design IoT applications in different domains and real time scenarios

UNIT 1 INTRODUCTION	**11 Hrs.**
Definition & Characteristics of IoT, Technology evolution, IoT Architectures, Physical and Logical Design of IoT, IoT Functional Blocks, IoT Communication Models- Publish-subscribe, Push-Pull, exclusive-pair models, Resource management, IoT Security, Current challenges and issues.	
UNIT 2 IoT DEVICES AND TECNOLOGIES	**12 Hrs.**
Introduction to various types of sensors, Working principles of sensors, Actuators, Control Units-Microcontrollers and Microprocessors, I/O interfaces, Communication Technologies for IoT, RFID, IoT Enabling Technologies – AI, Bigdata Analytics, Cloud Computing, and Embedded Systems.	
UNIT 3 INDUSTRIAL IoT	**9 Hrs.**
Introduction to Industrial IoT, Role of Industrial Internet of Things (IIoT), Difference between IoT and IIoT, Business Models, IoT-based real-time control systems and Human-Machine Interface, Supervisory Control & Data Acquisition (SCADA), Case studies on Smart Factories with IIoT.	
UNIT 4 IoT SYSTEMS AND APPLICATIONS	**10 Hrs.**
Applied Internet of Things, Smart Cities, Smart Agriculture, Intelligent Transportation Systems, IoT Applications for Smart Health Care, Home Automation, Environment Monitoring, and Surveillance, Case Studies Illustrating IoT Design.	

Max. 42 Hrs.

COURSE OUTCOMES

On completion of the course, students will be able to

CO1	:	Remember the Internet of Things and its technological aspects.
CO2	:	Comprehend advanced IoT devices and technologies for different IoT applications.
CO3	:	Apply the knowledge of control units and communication technologies in designing IoT systems.
CO4	:	Analyse the Industrial systems with IoT components.
CO5	:	Evaluate efficiency trade-offs among alternative models for an efficient IoT application design
CO6	:	Design different IoT systems for various real-life problems and application domains

TEXT/REFERENCE BOOKS

3. Pethuru Raj and Anupama C. Raman. The Internet of Things: Enabling Technologies, Platforms, and Use Cases. CRC Press.
4. Adrian McEwen. Designing the Internet of Things. Wiley publications.
5. Vijay Madisetti, Arshdeep Bahga. İnternet of Things: A Hands-on Approach. University Press.
6. Raj Kamal. Internet of Things: Architecture and Design. McGraw Hill

24ICxxxT					Modern Wireless Communications					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- Understand wireless communication evolution, cellular concepts, and system design.
- Analyze radio wave propagation and multipath fading in wireless systems.
- Evaluate performance of wireless communication systems in the presence of fading.
- Explore diversity, multiplexing, and access techniques for enhanced wireless communication.

UNIT 1: Introduction to Wireless Communications	**09 Hrs.**
Evaluation of Wireless Communications (2G to 5G); Cellular Concept: System Design Fundamentals, Frequency Reuse, Handoff, Cochannel Interference and System Capacity	
UNIT 2: Mobile Radio Propagation	**12 Hrs.**
Path Loss and Shadowing: Radio Wave Propagation, Transmit & Receive Signal Models, Free-Space Path Loss, Shadow Fading; Statistical Multipath Channel Models: Narrowband and Wideband Fading, Rayleigh and Rician Fading Distribution; Channel Equalizer; Performance Analysis over Fading Channel: Outage Probability, Average Probability of Error, Capacity	
UNIT 3: Space-Time Diversity and Multiplexing	**11 Hrs.**
SIMO, MISO and MIMO Models; Receiver Diversity: Selection Combining, Equal Gain Combining, Maximal-Ratio Combining, Transmit Diversity: Alamouti Space-Time Coding Scheme. Spatial Multiplexing.	
UNIT 4: Multiple Access Techniques	**10 Hrs.**
Spread Spectrum Communication: Spread Spectrum Principles, Direct Sequence Spread Spectrum, Frequency Hoping Spread Spectrum; Multicarrier Modulation Schemes: Orthogonal Frequency Division Multiplexing (OFDM), Cyclic Prefix, Case Study: The IEEE 802.11a Wireless LAN Standard; Introduction to Next Generation Wireless Communication Technologies.	

Total: 42 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Comprehend wireless system evolution, cellular concepts, and design fundamentals.
CO2: Analyze radio wave propagation, path loss, fading models, and multipath effects.
CO3: Evaluate wireless communication system performance under fading conditions.
CO4: Apply diversity, multiplexing, and access techniques for enhanced communication.
CO5: Understand multiple antenna systems to improve communication performance.
CO6: Utilize various multiple access techniques in modern wireless communication systems.

TEXT/REFERENCE BOOKS

25. A. Goldsmith, “Wireless Communications,” Cambridge University Press, 2005.
26. D. Tse and P. Viswanath, “Fundamentals of Wireless Communications,” Cambridge University Press, 2005.
27. T. S. Rappaport, “Wireless Communications: Principles and Practice,” 2nd Ed., Prentice Hall, 2002.
28. Yifei Yuan, Zhifeng Yuan, “5G New Radio Non-Orthogonal Multiple Access”, CRC Press, 2023.

<Course Code>					BLOCKCHAIN TECHNOLOGY					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the concepts of block chain technology
2. To understand the consensus and hyper ledger fabric in blockchain technology.
3. To develop blockchain based solutions and write smart contract.

UNIT I: BLOCKCHAIN INTRODUCTION	10 Hrs.
History of blockchain, Growth of blockchain technology, Types of blockchain, Digital Money to Distributed Ledgers, Smart contracts, Blockchain Architecture, Use cases, Blockchain vs shared Database, Introduction to cryptocurrencies, Applications.	
UNIT II: BITCOIN	**11 Hrs.**
Concept of Double Spending, Hashing, Proof of work. Introducing Bitcoin, Bitcoin digital keys and addresses, Transactions, Blockchain mining. Bitcoin Network and payments, Bitcoin network, Wallets, Bitcoin payments, Innovation in Bitcoin, Bitcoin Clients and APIs. Alternative Coins. Limitations of Bitcoin.	
UNIT III: BLOCKCHAIN PLATFORMS	**11 Hrs.**
Ethereum, Hyperledger, IOTA, EOS, Multichain, Bigchain, etc. Advantages and Disadvantages, Ethereumvs Bitcoin, Design a new blockchain, Potential for disruption, Design a distributed application, Blockchain applications.	
UNIT IV: PRIVACY, SECURITY ISSUES IN BLOCKCHAIN	**10 Hrs.**
Pseudo-anonymity vs. anonymity, Zcash and Zk-SNARKS for anonymity preservation, attacks on Blockchains – such as Sybil attacks, selfish mining and 51% attacks; Sharding based consensus algorithms. Case Studies of blockchain in Finance, Management and Government	
	TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Understand the types, benefits and limitation of blockchain.
CO2	:	Understand the concepts of Bitcoin and their usage.
CO3	:	Apply security features in blockchain technologies.
CO4	:	Analyze distributed computing and cryptography related to blockchain
CO5	:	Evaluate Use smart contract in real world applications.
CO6	:	Create the smart contracts and Blockchain features for outside of currencies.

TEXT/REFERENCE BOOKS

1. Narayanan, Bonneau, Felten, Miller and Goldfeder, “Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction”, Princeton University Press.
2. Josh Thompson, ‘Blockchain: The Blockchain for Beginnings, Guild to Blockchain Technology and Blockchain Programming’, Create Space Independent Publishing Platform, 2017.
3. Imran Bashir, “Mastering Blockchain: Distributed ledger technology, decentralization, and smart contracts explained”, Packt Publishing.
4. Merunas Grincalaitis, “Mastering Ethereum: Implement Advanced Blockchain Applications Using Ethereum-supported Tools, Services, and Protocols”, Packt Publishing.
5. Mastering Block chain - Distributed ledgers, decentralization and smart contracts explained, Author- Imran Bashir, Packt Publishing Ltd, Second Edition, ISBN 978-1-78712-544-5, 2017
6. Mark Gates, “Block chain: Ultimate guide to understanding block chain, bit coin, crypto currencies, smart contracts and the future of money”, Wise Fox Publishing and Mark Gates 2017.
7. Salman Baset, Luc Desrosiers, Nitin Gaur, Petr Novotny, Anthony O'Dowd, Venkatraman Ramakrishna, “Hands-On Block chain with Hyper ledger: Building decentralized applications with Hyperledger Fabric and Composer”, 2018.
8. Bahga, Vijay Madisetti, “Block chain Applications: A Hands-On Approach”, Arshdeep Bahga, Vijay Madisetti publishers 2017.

24ICXXXT					Mobile Application Development					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- Learn fundamentals of Mobile development.
- Understand various technologies and business trends impacting mobile applications.
- Apply a deep knowledge of mobile devices, features, architecture and android functionality.

UNIT 1 Overview of Android Development	**9 Hrs.**
Android OS architecture, Introducing development framework, Dalvik virtual machine – DVM, android virtual device and SDK manager, developing and executing the first android application, android activities- creating an activity, managing the lifecycle.	
UNIT 2: User Interface Design and Interaction in Android Development	**10 Hrs.**
Working with Views- Text, Edit Text, Button, Radio Button, Checkbox, Image Button, Toggle Button, Rating Bar, Working with View Groups- Linear Layout, Relative Layout, Constraint Layout, Scroll View, Table, Frame, Table with Action Bar, Binding Data with the Adapter View Class- List View, Spinner, Gallery View, Creating Menus & Dialogs	
UNIT 3 : Android Navigation and Fragment Management	**11 Hrs.**
Intent Objects, Intent Filters, Linking the Activities Using Intent, Obtaining Results from Intent, Passing Data Using an Intent Object, Fragments- Fragment Implementation, Finding Fragments, Adding, Removing, and Replacing Fragments	
UNIT 4 : Data Management and Connectivity in Android	**12 Hrs.**
Understanding SQLite Database and Data Types, Working with Cursors and Content Values in SQLite, Using SQLiteOpenHelper for Database Management, Adding, Updating, and Deleting Content in SQLite, Integrating XML & JSON Based Web Services, Introduction to Firebase for Android and Firebase Connectivity	

Total: 42Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 – Understand various technologies and business trends impacting mobile applications

CO2 – Apply a deep knowledge of mobile device, features, architecture and android functionality

CO3 – To explore and learn flutter to develop real time mobile applications

CO4 – Analyze and implement frameworks, database and design patterns in Mobile applications

CO5 – Learn to build features that have data persistence and data communications

CO6 – Create a mobile application using the Android programming language.

TEXT/REFERENCE BOOKS

1. Android Developer Tools Essentials by Mike Wolfson - O'Reilly Media Publications.
2. Bill Phillips, Chris Stewart, Brian Hardy, and Kristin Marsicano, Android Programming: The Big Nerd Ranch Guide, Big Nerd Ranch LLC, 2nd edition.
3. Learn Java for Android Development, 2nd Edition - Jeff Friesen - Apress Publications